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PINNGraPE

Physics-Informed Neural Network for Gravitational- wave Parameter Estimation

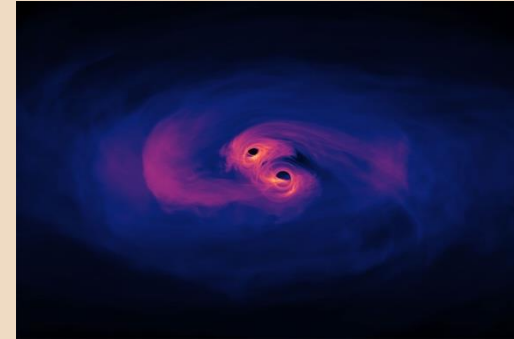
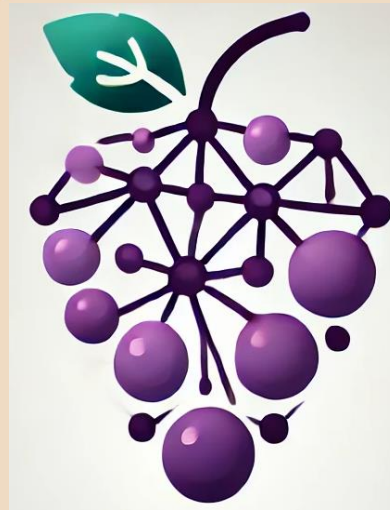
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May 15th, 2025

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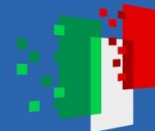




Introduction

- PINNGraPE is a ML algorithm which performs PE on CBC burst unmodeled signals.
- Based on Physics Informed Neural Networks.

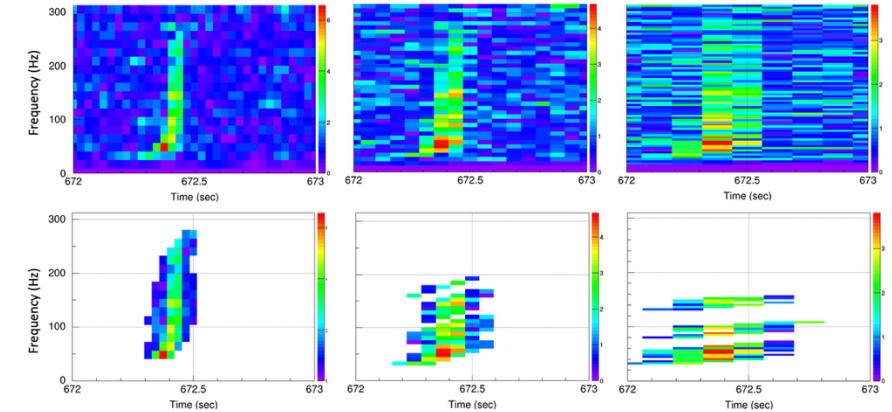
[2]



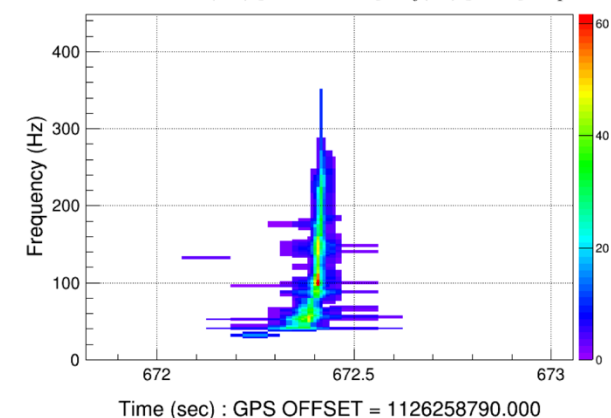
Introduction: cWB

- cWB is an unmodeled pipeline used by LVK collaboration [1].
- **Unmodeled**: based on pure coherence between energy pixels in the frequency domain.
- **No theoretical background** needed to detect sources:
 - **Pro**: search for **new signals**.
 - **Con**: when detecting CBCs, the **PE** on the source, i.e. the chirp mass estimation, is **only given by a fit** on the likelihood plot.

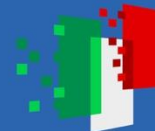
Let's fix it!



Likelihood 641 - $dt(ms)$ [7.8125:250] - $df(hz)$ [2:64] - $npix$ 131

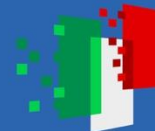


[1]



Introduction: PINNs

- Physics-Informed Neural Networks are a ML technique developed to **exploit, discover or solve differential equations** as physical laws behind the data [2, 3].
- The base of the physical information is the **loss function**: the physics is implemented inside the loss itself in form of PDE and auxiliary physics relations.



Main goal for PINNGraPE

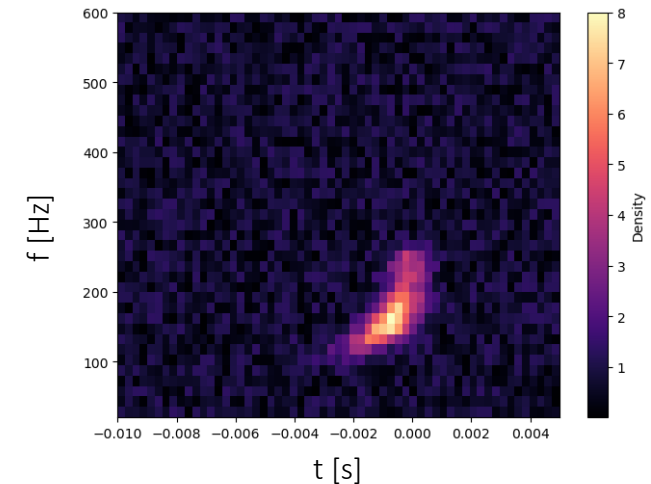
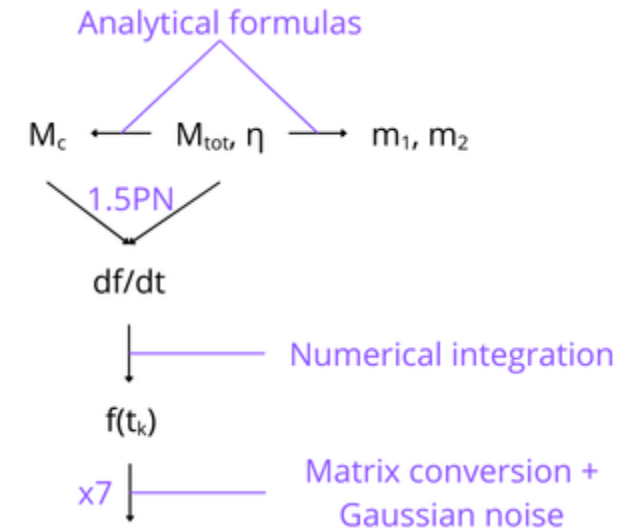
To infer the two BH masses m_1 and m_2 from a dataset of simulated burst CBCs signals.

Dataset: 10^4 simulated signals. For every signal:

- 7 spectrogram-like images
- training: 70%, validation: 20%, test: 10%.
- Stored values for target:
 - $\{m_1, m_2, M_c, M_{\text{tot}}, \eta, \{df/dt\}_k, \text{Newt}, \{df/dt\}_{k, 1.5\text{PN_corr}}, f_k\}$,
- 1.5PN formalism [4]:

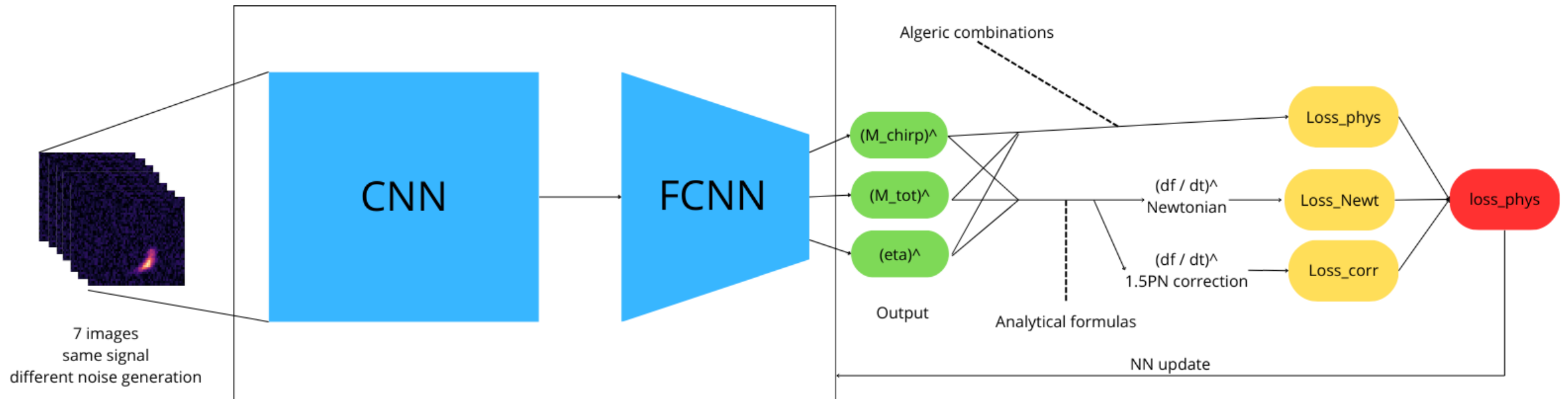
$$\varepsilon = \frac{GM_\odot}{c^3} \left(\frac{M_{\text{tot}}}{M_\odot} \right) \pi f$$

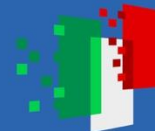
$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_\odot}{c^3} \frac{M_{\text{tot}}}{M_\odot} \eta^{3/5} \right)^{5/3} f^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) \varepsilon^{2/3} + 4\pi \varepsilon \right]$$





Architecture





Loss function

$$\mathcal{L} = \mathcal{L}_{Newt} + \mathcal{L}_{1.5PN} + \mathcal{L}_{phys} = \frac{\beta_1}{B} \sum_{i=1}^B \left(\frac{d\hat{f}_i}{dt} - \frac{df_i}{dt} \right)_{Newt}^2 + \frac{\beta_2}{B} \sum_{i=1}^B \left(\frac{d\hat{f}_i}{dt} - \frac{df_i}{dt} \right)_{1.5PN-Newt}^2 + \mathcal{L}_{phys}$$

$$\mathcal{L}_{phys} = \frac{\beta_3}{B} \sum_{i=1}^B \left(\hat{M}_{tot,i} \hat{\eta}_i^{3/5} - \hat{\mathcal{M}}_i \right)^2 + \frac{\beta_4}{B} \sum_{i=1}^B \left(\hat{\mathcal{M}}_i \hat{\eta}_i^{-3/5} - \hat{M}_{tot,i} \right)^2 + \frac{\beta_5}{B} \sum_{i=1}^B \left(\left(\frac{\hat{\mathcal{M}}_i}{\hat{M}_{tot,i}} \right)^{3/5} - \hat{\eta}_i \right)^2$$

- Main terms: df/dt ones.
Newtonian one depends only on M_c , while the correction one depends on M_{tot} and η .
- Physical loss terms exploit physical redundancy between M_c , M_{tot} and η .
Physical redundancy explained in detail in Di Clemente et al. [5] and in a new paper in the writing stage.
- Notice: no direct output vs target term.



Best run hyperparameters

- $\beta_1 = 10^{-6}$, $\beta_2 = 10^4$, $\beta_3 = \beta_4 = \beta_5 = 1$.
- Optimizer: AdamW
- Scheduler: ReduceLROnPlateau(factor = 0.25, patience = 66)
- Break for $lr < 10^{-6}$.

- Final train loss value: 0.0951434
- Final validation loss value: 0.4111404

- Time needed (GPU):
 - Train: 947 s
 - Test: ~2 ms for 1 signal



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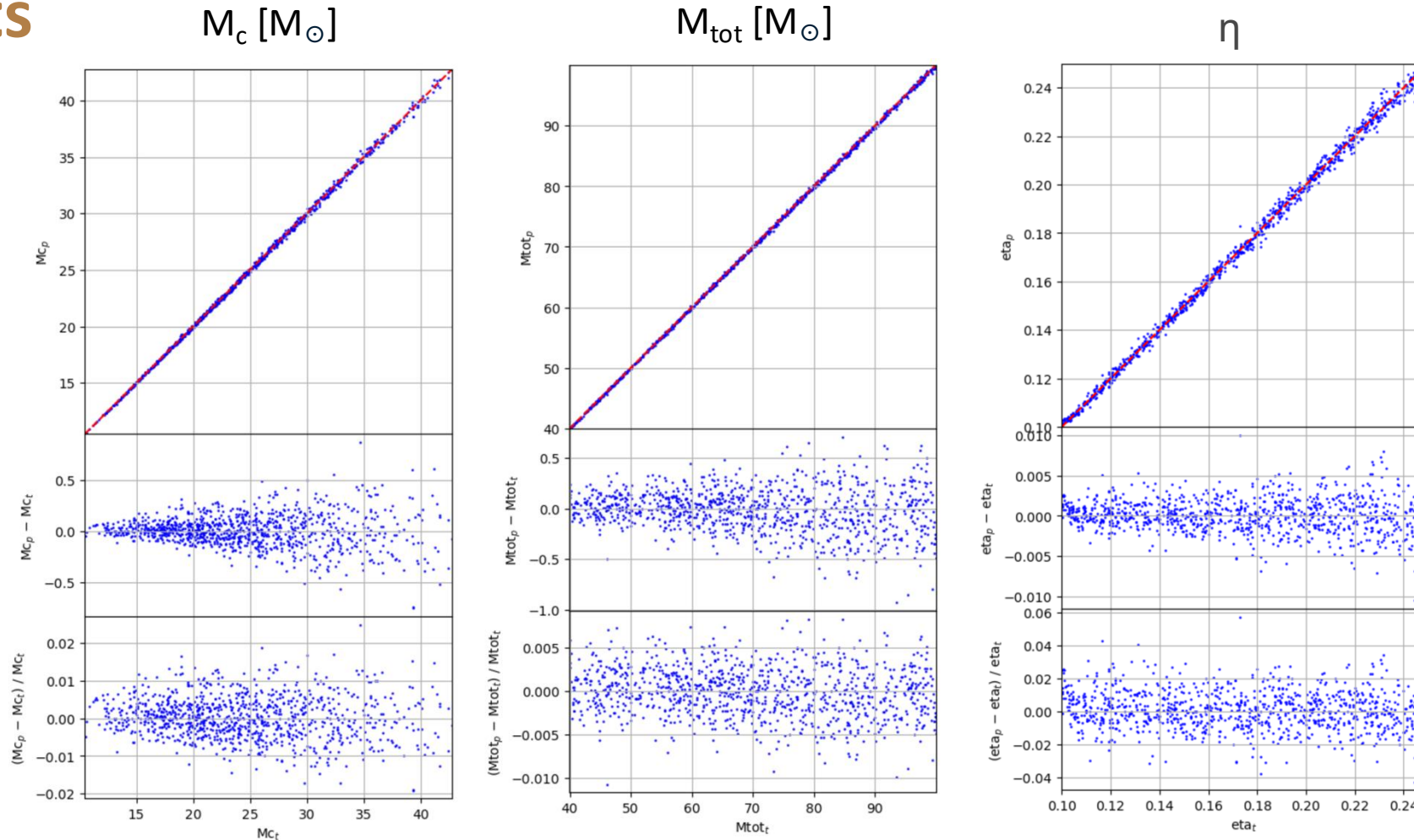
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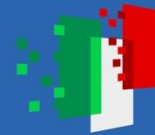


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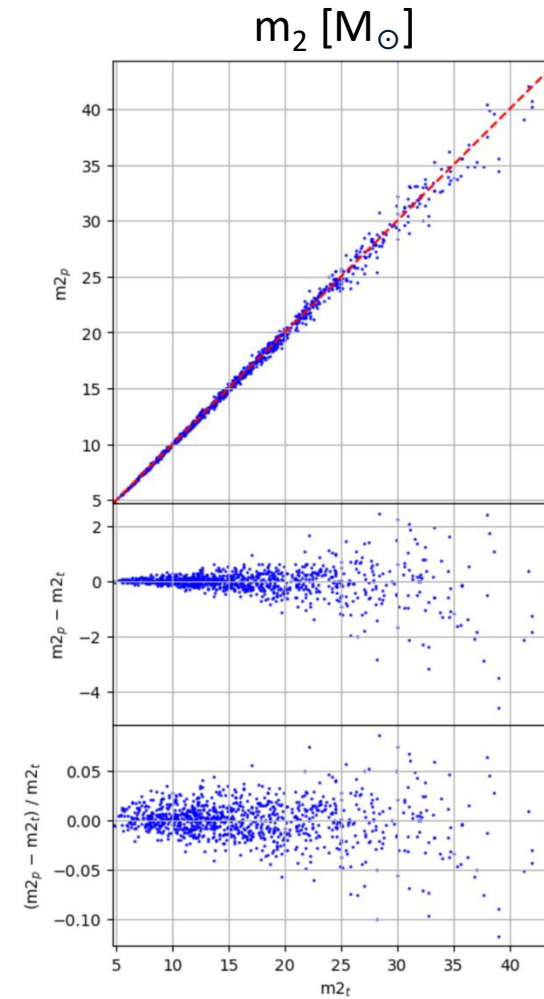
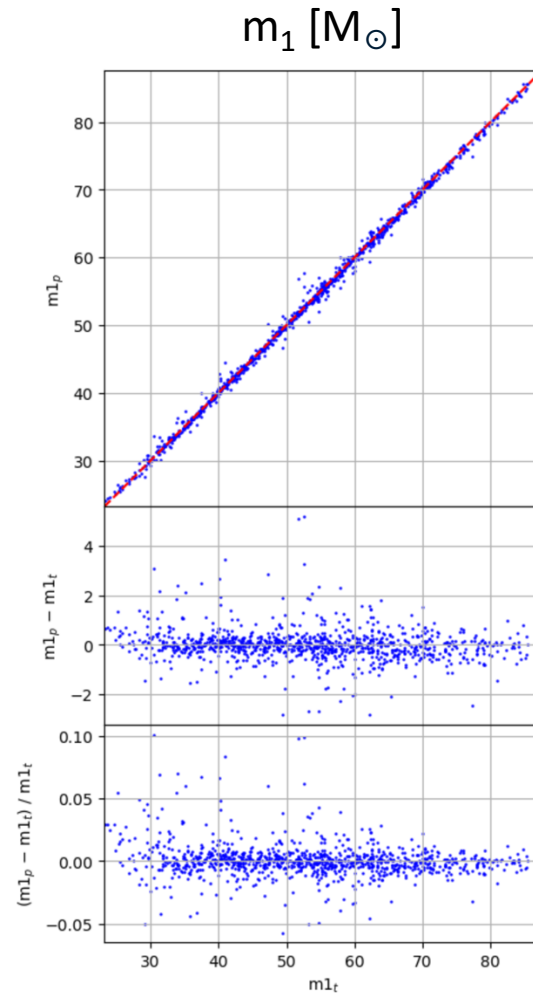


Results





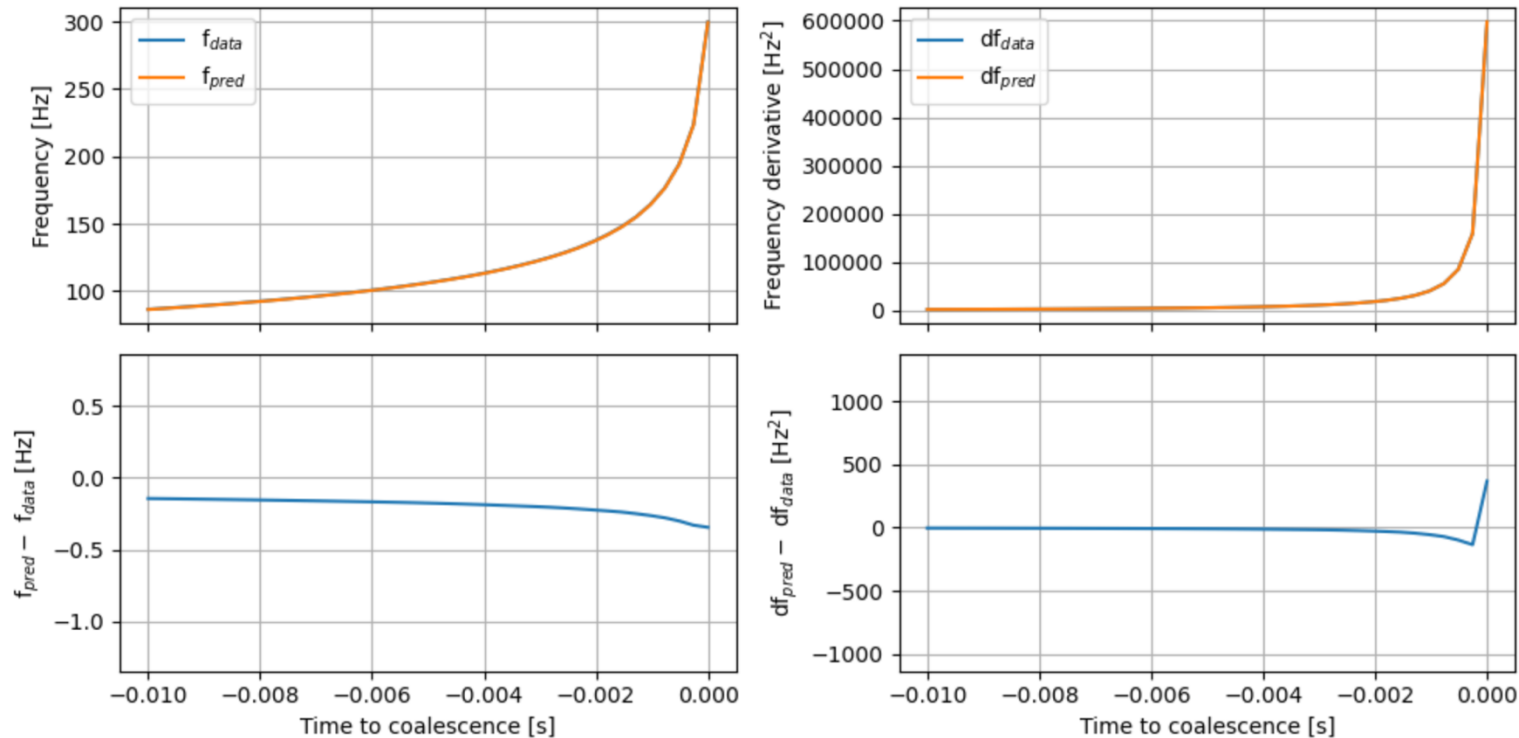
Results





Results

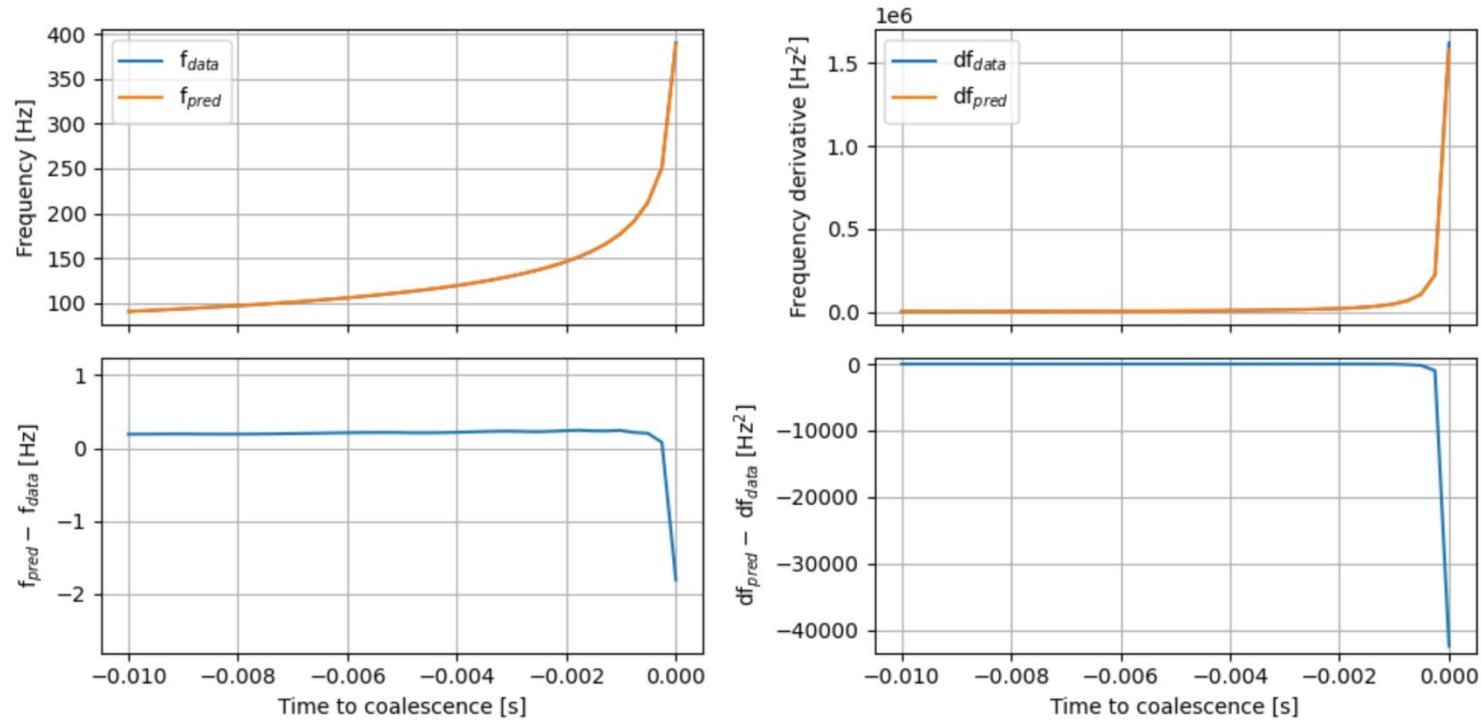
Variables	Target [M _{sun}]	Predicted [M _{sun}]	Difference [M _{sun}]	Relative Difference []
m1	62.2199211	62.2788620	0.0589409	0.0009473
m2	14.1337042	14.1629267	0.0292225	0.0020676
Mc	24.5439987	24.6159897	0.0719910	0.0029331
Mtot	76.3536224	76.4417877	0.0881653	0.0011547
eta	0.1508433	0.1509496	0.0001063	0.0007045





Results

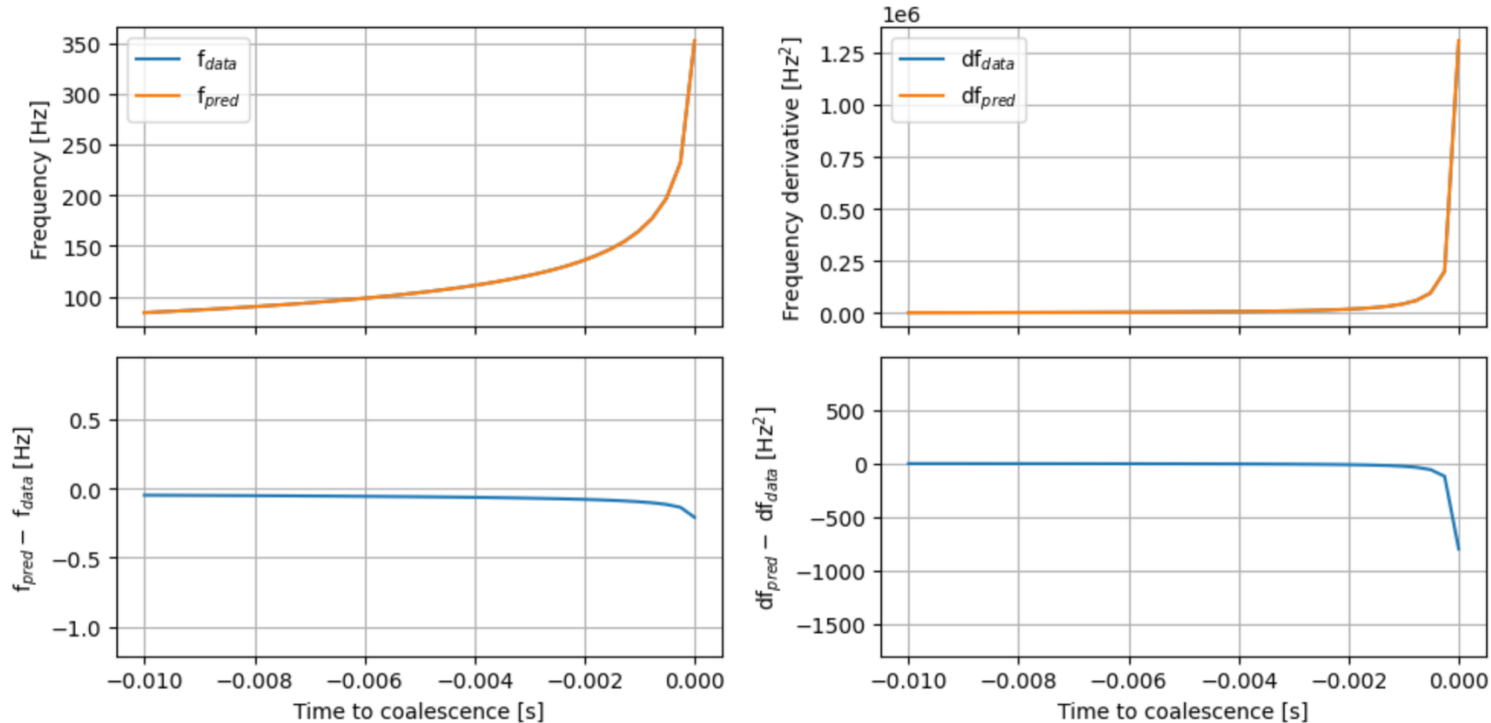
Variables	Target [M _{sun}]	Predicted [M _{sun}]	Difference [M _{sun}]	Relative Difference []
m1	30.4501419	33.5333672	3.0832253	0.1012549
m2	28.1909962	25.3806343	-2.8103619	-0.0996900
Mc	25.5023003	25.3342171	-0.1680832	-0.0065909
Mtot	58.6411400	58.9140015	0.2728615	0.0046531
eta	0.2496290	0.2452125	-0.0044165	-0.0176921





Results – test with GW150914 like masses

	Target [M _{sun}]	Predicted [M _{sun}]	Difference [M _{sun}]	Relative difference []
m1	36.0000000	35.9720192	-0.0279808	-0.0007772
m2	29.0000000	29.0668106	0.0668106	0.0023038
Mc	28.0955563	28.1226387	0.0270824	0.0009639
Mtot	65.0000000	65.0388336	0.0388336	0.0005974
eta	0.2471006	0.2471820	0.0000814	0.0003293





Conclusions and future steps

- Main goal reached:
 m_1 and m_2 are inferred within an error $< 5\%$.
- PINNs really represents a promising approach for achieving fast PE for unmodeled signals.
- Next steps:
 - We are almost ready for cWB implementation (stay tuned!).
 - Study on the error estimate on a new incoming signal.
 - Increase the parameters to estimate: add further PN terms.



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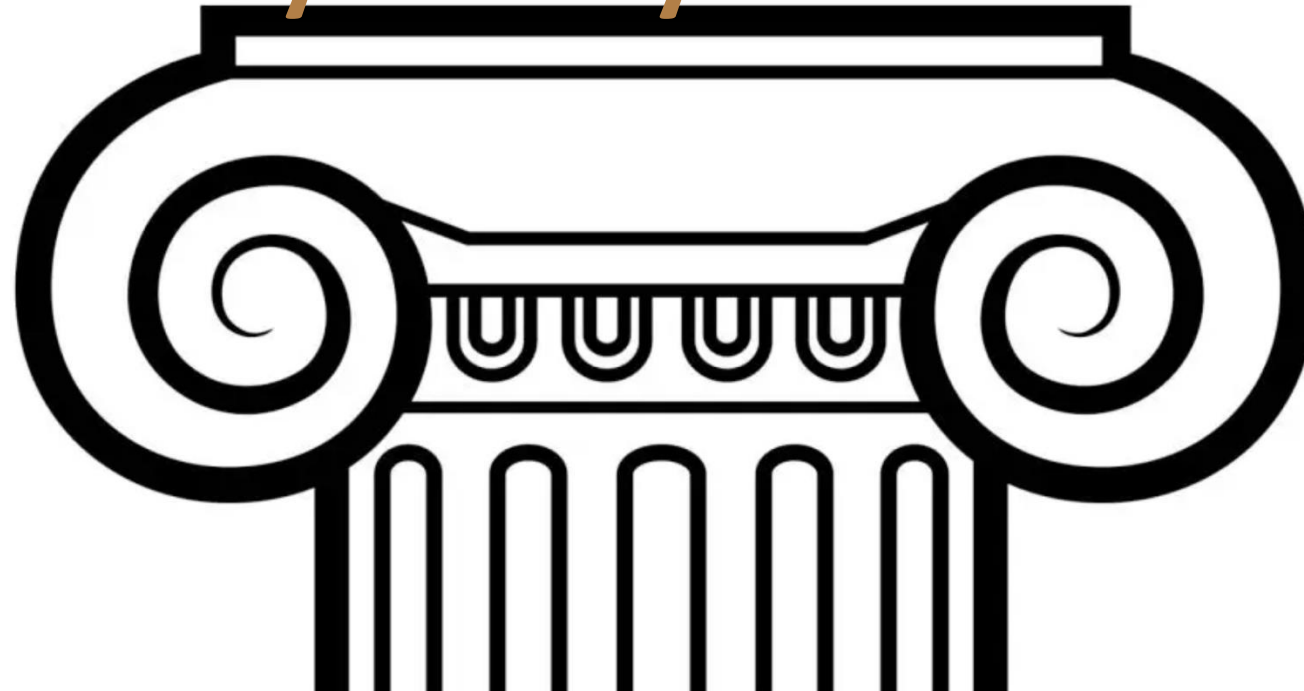
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Thank you for your attention!





References

- [1] Drago, M., Klimenko, S., Lazzaro, C., Milotti, E., Mitselmakher, G., Nacula, V., O'Brian, B., Prodi, G. A., Salemi, F., Szczepanczyk, M., Tiwari, S., Tiwari, V., V. G., Vedovato, G., and Yakushin, I. (2021). coherent waveburst, a pipeline for unmodeled gravitational-wave data analysis. *SoftwareX*, 14:100678.
- [2] Raissi, M., Perdikaris, P., and Karniadakis, G. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707.
- [3] Nascimento, R. G., Fricke, K., and Viana, F. A. (2020). A tutorial on solving ordinary differential equations using python and hybrid physics-informed neural network. *Engineering Applications of Artificial Intelligence*, 96:103996.
- [4] Cutler, C. and Flanagan, E. E. (1994). Gravitational waves from merging compact binaries: How accurately can one extract the binary's parameters from the inspiral waveform? *Physical Review D*, 49(6):2658–2697.
- [5] F. Di Clemente, M. Scialpi, and M. Bejger, “Explainable autoencoder for neutron star dense matter parameter estimation”, 2025.



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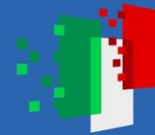
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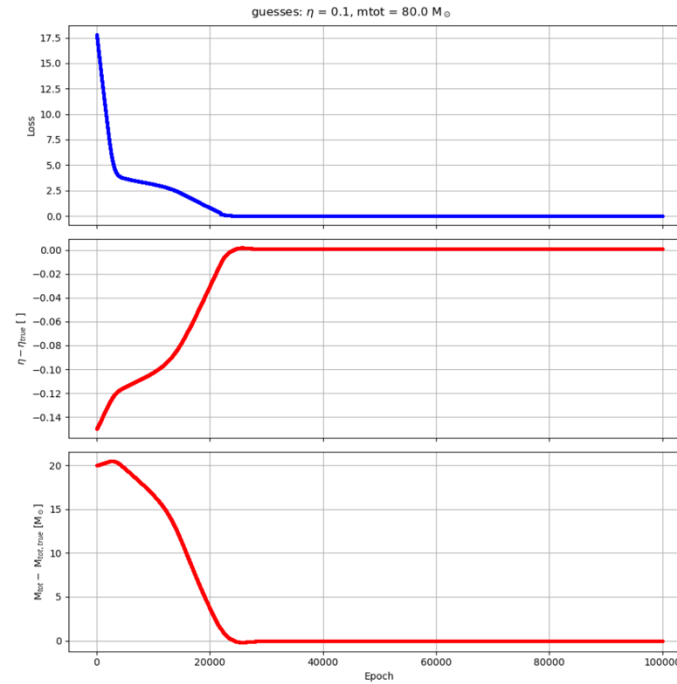
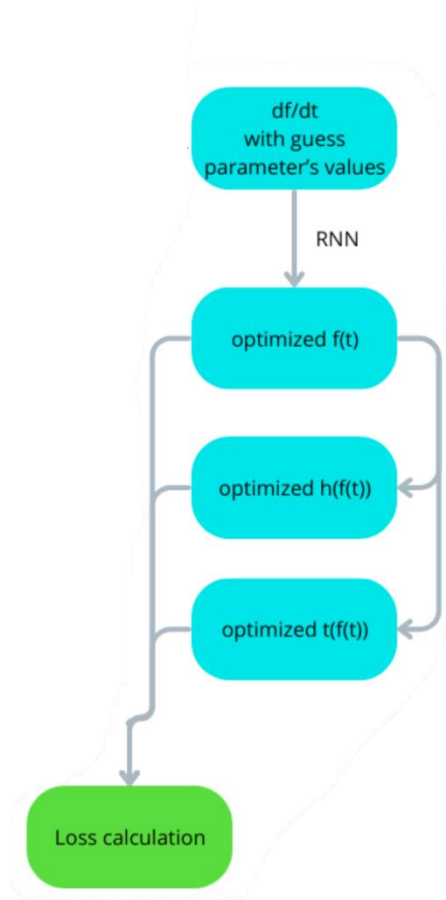
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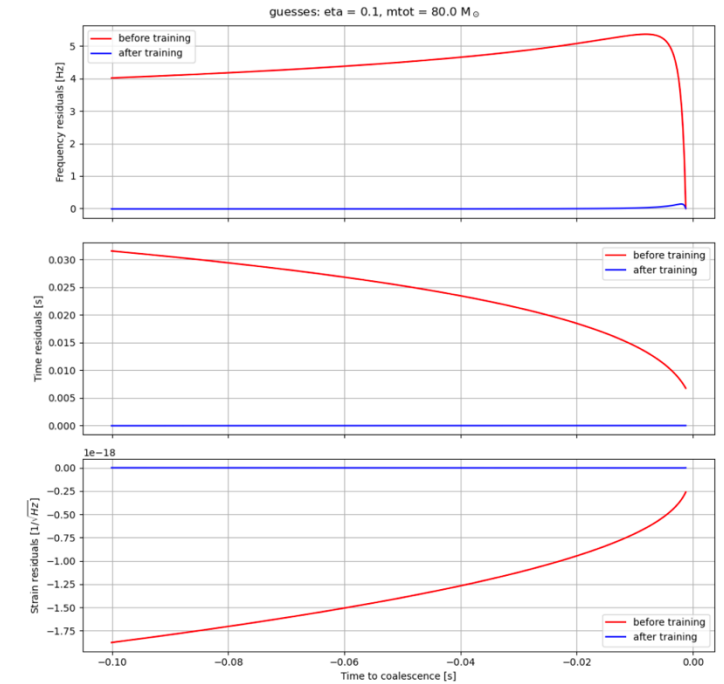
Backup slides



RNN version



- Final loss value: $2.385 \cdot 10^{-2}$.
- AdamW optimizer
- Total time: -1h on CPU (4000 epochs).



	η	$M_{tot} [M_{\odot}]$
Guess	0.1000	80.00
Target	0.2500	60.00
Inferred	0.2507	59.95



1.5PN formalism [4]

$$M_{\text{tot}} = m_1 + m_2$$

$$\eta = \frac{m_1 m_2}{(m_1 + m_2)^2} = \frac{m_1 m_2}{M_{\text{tot}}^2}$$

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} = M_{\text{tot}} \eta^{3/5}$$

$$\varepsilon = \frac{GM_{\odot}}{c^3} \left(\frac{M_{\text{tot}}}{M_{\odot}} \right) \pi f$$

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_{\odot}}{c^3} \frac{M_{\text{tot}}}{M_{\odot}} \eta^{3/5} \right)^{5/3} f^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) \varepsilon^{2/3} + 4\pi \varepsilon \right]$$



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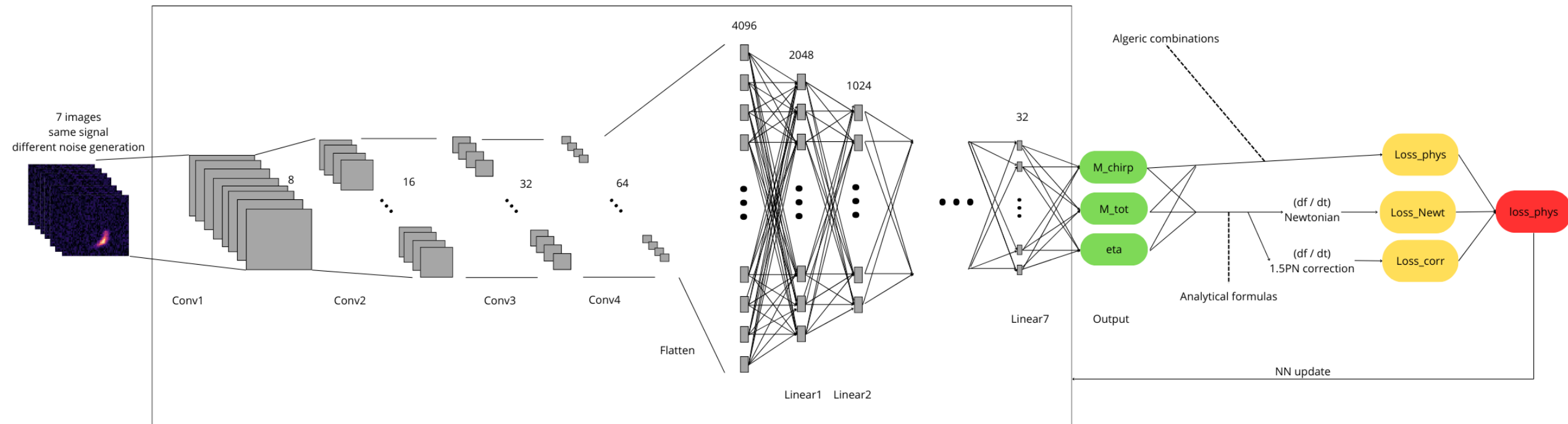
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Architecture





PINNs theory

$$u_t + \mathcal{N}[u; \lambda] = 0, \quad x \in \Omega, \quad t \in [0, T],$$

$$f := u_t + \mathcal{N}[u],$$

$$MSE = MSE_u + MSE_f,$$

where

$$MSE_u = \frac{1}{N_u} \sum_{i=1}^{N_u} |u(t_u^i, x_u^i) - u^i|^2,$$

and

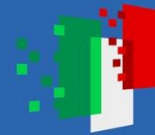
$$MSE_f = \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_f^i, x_f^i)|^2.$$

Task: find an approximation for u (= solve the differential equation) and optimize λ .

Data: (u^i, t^i, x^i) .

How: approximate u with a NN and minimize the loss.

Why this should work? Since u is a NN, f is a PINN: they share the same parameters.



PINNs theory

